Comparison of Liquid Thermal and Electrocatalysis for the Selective Oxidation of Ethylene Glycol over Cobalt Spinel Oxides

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Introduction

Ethylene glycol (EG) is a potentially green feedstock that can be oxidized to industrially relevant products such as glycolic acid (GA) used in packaging materials and adhesives[1]. Highly active and selective catalysts are reported in liquid thermal and electrocatalytic oxidation of EG[2-4]. However, the preparation of low-cost and highly stable catalysts remains a major challenge. Cobalt spinel oxides have been studied in liquid thermal and electrocatalytic reactions such as CO or alcohol oxidation because of their high poisoning resistance, thermal stability and low costs[5]. The robustness of Co_3O_4 allows the comparison of liquid thermal oxidation (LTO) and electrooxidation in alkaline conditions. Herein, cobalt spinels were synthesized by spray flame synthesis (SFS), co-precipitation and hard templating using SBA-15 for the selective oxidation of ethylene glycol to glycolic acid.

Results and Discussion

The synthesis method of Co_3O_4 shows a clear effect on the yield of GA under mild conditions. For LTO, SFS results in the highest yield and low formation of oxalic acid (Fig.1a). Ethylene glycol oxidation reaction (EGOR) demonstrates the oxidation capacity to GA with larger production of formate and oxalate finally reaching the oxygen evolution reaction regime at higher potential. The LSV curves suggest the

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oxidation of Co^{2+} to Co^{3+} with a higher EGOR overpotential for the SFS sample (Fig.1b-c). Solid-state EPR shown in Figure 1d suggests a major dislocation of Co^{2+} in the Co_3O_4 prepared by co-precipitation, which may explain the lower performance in LTO, since the altered structure can affect the pairs of Co^{3+} cations, assumed as the active sites. Thus, the synthesis method significantly influences the catalytic properties in the thermal and the anodic oxidation of ethylene glycol, which seem to occur on the same active sites yielding glycolate as main product.

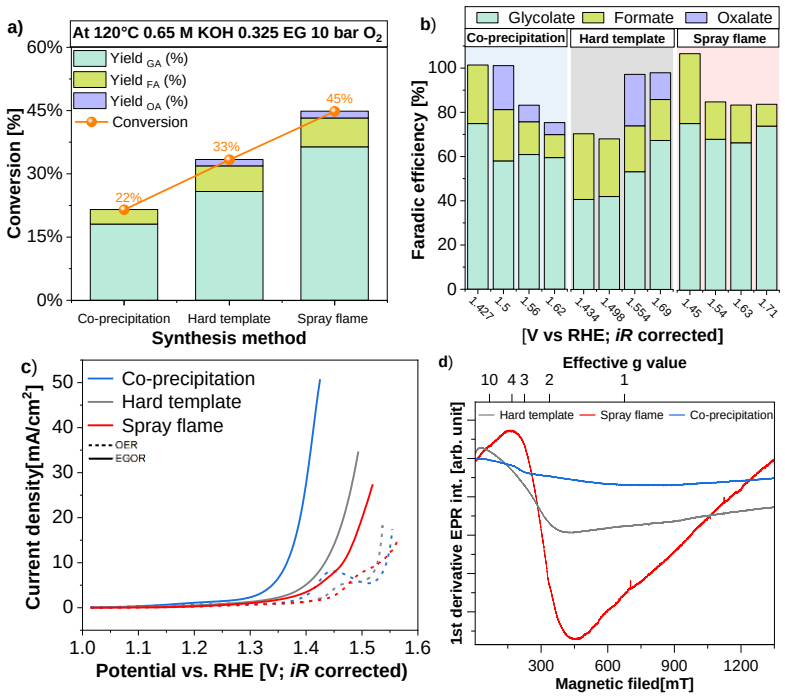

Figure 1. Effect of the synthesis method on the conversion and product yields in the **a)** liquid thermal oxidation **b)** electrocatalytic oxidation of ethylene glycol. **c)** LSV curves. **d)** Solid-state EPR spectra of the three differently synthesized Co_3O_4 catalysts.

Conclusions

LTO showed higher yields of 35% glycolic acid for the SFS method, whereas Co_3O_4 prepared by co-precipitation had the lowest overpotential for the EGOR, accompanied by a loss in faradic efficiency with increasing potential.

Significance

The selective oxidation of ethylene glycol allows the comparison of liquid thermal and electrocatalytic oxidation. The wet-chemical and the gas-phase synthesis of Co_3O_4 provided low-cost, resistant and active catalysts. Hence, our work provides valuable insight into the similarities as well as differences of liquid thermal and anodic oxidation of ethylene glycol to glycolic acid over differently synthesized Co_3O_4 catalysts aiming at structure-activity correlations.